

Assessment 2 Applied NLP Project

Robustness Benchmarking on AG News

This report summarizes an end-to-end NLP project built around the AG News topic classification dataset. The main goal is not only to achieve good accuracy on clean text, but also to compare how different models behave when the input is deliberately corrupted with realistic surface-level noise. I compare two classical sparse-feature baselines with a pretrained DistilBERT checkpoint that was already fine-tuned on AG News.

1. Problem Definition

AG News is a four-class news topic classification task. Each short article snippet must be assigned to one of four labels: World, Sports, Business, or Sci/Tech. This is a natural NLP problem because the model must interpret free text and map it to a semantic category.

The project is interesting for two reasons. First, the dataset is a simple and well-known benchmark for text classification, so it is easy to understand and evaluate. Second, the benchmark can be extended into a robustness study by testing how models react to typos, missing words, and vowel masking. That makes the project more realistic than a standard clean-text classification exercise, because many real text systems must cope with noisy user-generated language.

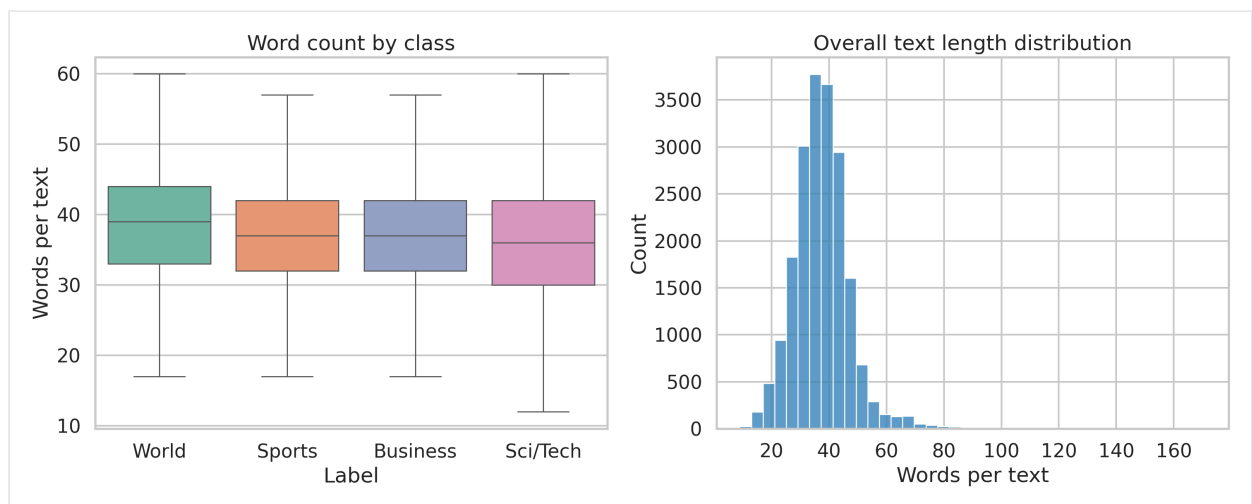
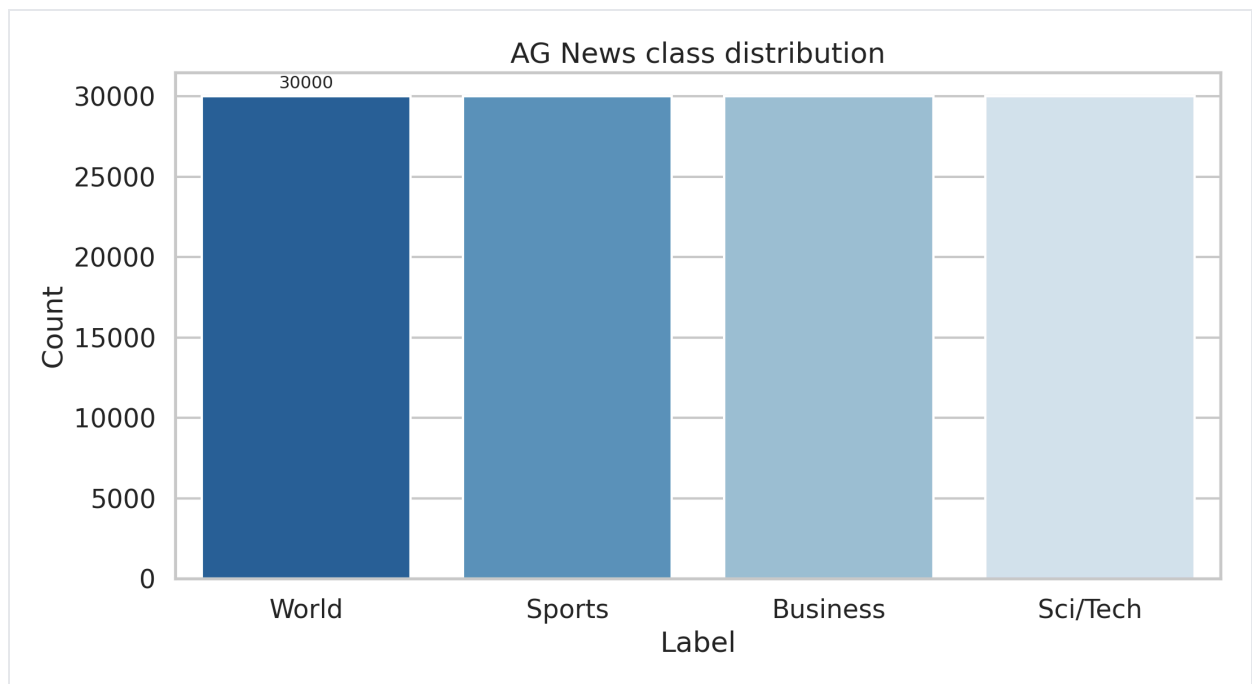
The main question in this report is: which model performs best on clean text, and which model stays strongest when the text becomes noisy?

2. Dataset Understanding and EDA

I used the `ag_news` dataset from Hugging Face. The dataset is balanced across all four labels, with 120,000 training examples and 7,600 test examples. Each class has exactly 30,000 training samples and 1,900 test samples. That balance is helpful because it makes accuracy a meaningful headline metric, while macro-F1 still lets us inspect class-level differences.

The articles are short and fairly consistent in length. In the training split, the average text length is about 37.8 words and 235.9 characters. The test split is almost identical, with an average of about 37.7 words and 234.7 characters. The length distribution is fairly tight, which suggests the task is driven more by topical cues than by long-document reasoning.

Sports snippets are slightly shorter on average than the other categories, while World and Business items are a little longer. That difference is small, so length alone would not be enough to solve the task. The class distribution and length plots below show the dataset structure clearly.



Overall, the EDA suggests a clean, balanced benchmark with limited obvious bias in class size. The main challenge is therefore not class imbalance, but the ability of the model to detect topical language from short snippets.

3. Preprocessing and Model Design

The preprocessing pipeline is intentionally lightweight. I only normalize whitespace and preserve the original text otherwise. This keeps the benchmark simple and avoids removing tokens that may be useful for topical classification.

I compare three models:

1. Word n-gram TF-IDF + Linear SVM

- Uses word n-grams with `ngram_range=(1, 2)`.
- Removes English stop words.
- Uses `min_df=2`, `max_df=0.95`, and sublinear TF scaling.
- This is a strong and interpretable sparse-text baseline.

2. Character n-gram TF-IDF + Linear SVM

- Uses character n-grams with `analyzer="char_wb"` and `ngram_range=(3, 5)`.
- Character features preserve subword shape, which makes them more tolerant of spelling variation and corrupted tokens.
- This model is expected to be more robust than the word model under typo-like noise.

3. Pretrained DistilBERT checkpoint fine-tuned on AG News

- I used a pretrained transformer model rather than training a transformer from scratch.
- The checkpoint is loaded with a Hugging Face tokenizer and run in batches.
- The notebook automatically uses CUDA when it is available, which matters because transformer inference is much more expensive than the sparse SVM baselines.

For robustness testing, I apply three synthetic perturbations only to the test set:

- **Typo noise:** random character swaps, deletions, or duplications inside longer words.
- **Word dropout:** random removal of tokens.
- **Vowel masking:** removal of vowels from selected words.

Each attack is tested at 5%, 10%, and 20% severity. These are surface-level perturbations, so they keep the underlying topic mostly intact while changing the text form. That makes them suitable for measuring brittleness rather than semantic drift.

4. Evaluation and Results

Because the dataset is balanced, accuracy and macro-F1 are very close. I used both, but macro-F1 is the most informative summary metric because it weights all four classes equally.

The clean and perturbed results are shown below.

Model	Clean Accuracy	Clean Macro-F1	Avg Perturbed Macro-F1	Worst Perturbed Macro-F1	Mean Macro-F1 Drop
transformer_distilbert	0.9479	0.9479	0.9407	0.9306	0.0073
char_ngram_svm	0.9272	0.9271	0.9224	0.9169	0.0048
word_ngram_svm	0.9222	0.9220	0.9169	0.9108	0.0051

The transformer is the strongest model in absolute terms. It has the highest clean score and it also remains the best model after perturbation. The clean macro-F1 of 0.9479 is about 2.1 points higher than the character SVM and about 2.6 points higher than the word SVM.

The most interesting robustness detail is that the character model loses slightly less performance relative to its clean baseline. Its mean macro-F1 drop is 0.0048, compared with 0.0051 for the word model and 0.0073 for the transformer. In other words, the transformer is the best performer overall, but the character baseline is the most stable classical model under these synthetic attacks.

The attack-level averages show a consistent pattern as well:

Attack	Average Macro-F1 Across Models
typo	0.9280
dropout	0.9266
vowel_mask	0.9254

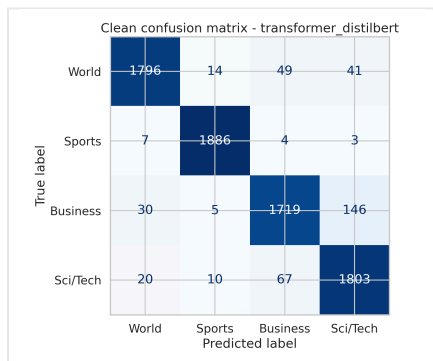
In this run, vowel masking was the most damaging perturbation overall, while typo noise was the least damaging on average. That is a useful reminder that not all "noise" affects models in the same way. Vowel masking removes a large amount of lexical identity, so it is especially disruptive for both sparse models and the transformer.

The class-level reports show a similar pattern across all three models. Sports is the easiest class, while Business is the hardest. For the transformer, Sports reaches an F1 score of 0.9887, whereas Business is lower at 0.9195. The same general ranking appears in the SVM baselines, with Business and Sci/Tech being the most confusable classes. This is likely because they share topical vocabulary such as company names, product terms, and market language.

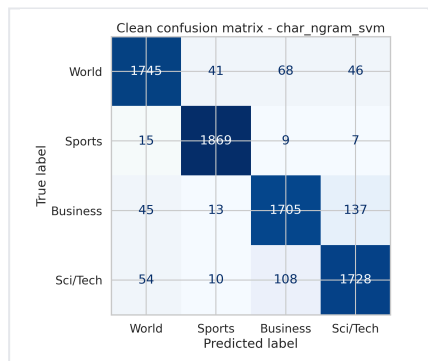
The generated robustness curves in `kaggle-outputs/` show the same overall ranking across all three perturbation types: the transformer stays highest in absolute performance, the character model is the best classical baseline, and the word model remains the weakest of the three.

Selected Visual Evidence

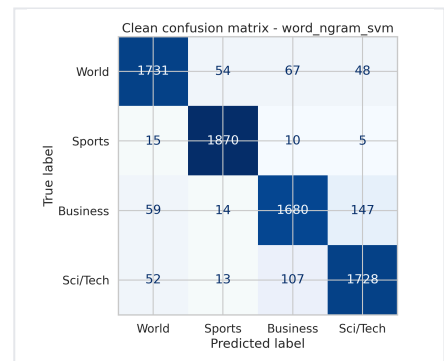
The first row below compares the clean confusion matrices for all three models. The transformer is strongest overall, while the classical baselines make more mistakes around the Business and Sci/Tech classes.



Transformer clean confusion matrix

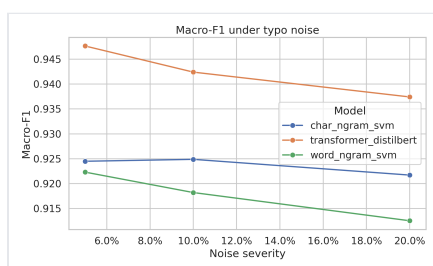


Character n-gram clean confusion matrix

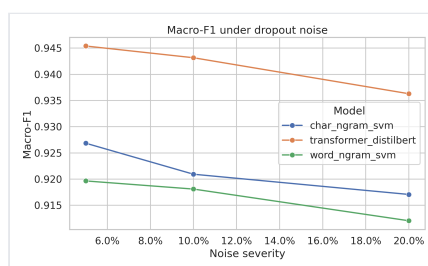


Word n-gram clean confusion matrix

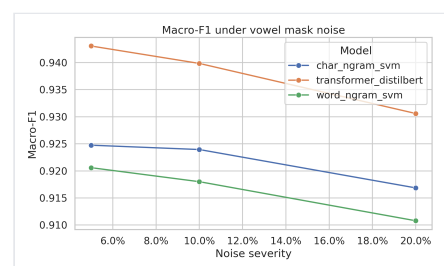
The second row shows how the benchmark behaves under the three synthetic perturbations. Vowel masking is the harshest attack in this run, and it affects every model more strongly than typo noise or dropout.



Typo robustness



Dropout robustness



Vowel masking robustness

These plots reinforce the summary table: the transformer remains the best model overall, the character SVM is the strongest classical baseline, and the word SVM trails slightly behind.

5. Discussion, Limitations, and Conclusion

This benchmark gives a clear answer to the main question. The pretrained DistilBERT checkpoint is the best model overall, both on clean text and under perturbation. If the goal is to maximize raw predictive performance, the transformer is the right choice.

At the same time, the classical models are still valuable. They are simpler, cheaper, and more interpretable, and the character-based model in particular is a strong robustness-oriented baseline. It does not beat the transformer in absolute terms, but it degrades more gently relative to its own clean score.

There are also several limitations to keep in mind. AG News is a balanced and relatively clean benchmark, so the task is easier than many real-world text classification problems. The perturbations are synthetic and only change surface form, so they do not capture deeper language variation such as paraphrasing, domain shift, or sarcasm. In addition, the transformer comparison uses an externally pretrained and already fine-tuned checkpoint, which is fair as a practical baseline but not the same as training a transformer from scratch on the dataset.

If I had more time, I would extend the benchmark in three ways. First, I would test stronger perturbations such as paraphrases or OCR-like corruption. Second, I would add a small calibration analysis to check whether confidence scores match correctness. Third, I would fine-tune a compact transformer myself and compare training time, accuracy, and robustness against the pretrained checkpoint.

In summary, the project shows that AG News is a clean classification task where a transformer gives the best overall performance, but character n-grams still provide a meaningful and robust classical baseline. The final takeaway is not just that one model wins, but that model choice depends on the trade-off between peak accuracy, robustness, and implementation simplicity.

References

- Hugging Face `ag_news` dataset.
- Hugging Face `textattack/distilbert-base-uncased-ag-news` checkpoint.
- scikit-learn for TF-IDF and Linear SVM baselines.
- PyTorch and Transformers for the pretrained model comparison.